192425037

EXP35:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <time.h>

int main() {
    char line[200], pt[200], ct[200], dt[200];
    int key[200], n = 0, i;

    printf("Enter plaintext: ");
    fgets(line, sizeof line, stdin);
    for (i = 0; line[i]; i++) if (isalpha(line[i])) pt[n++] = tolower(line[i]);
    pt[n] = '\0';

    /* key stream: one truly random shift (0..25) for every plaintext letter */
    srand((unsigned)time(NULL));
    for (i = 0; i < n; i++) key[i] = rand() % 26;

    printf("Key stream : ");
    for (i = 0; i < n; i++) printf("%d ", key[i]);
    printf("\n");

    for (i = 0; i < n; i++) ct[i] = (pt[i] - 'a' + key[i]) % 26 + 'A';
    ct[n] = '\0';
    printf("Ciphertext : %s\n", ct);

    for (i = 0; i < n; i++) dt[i] = (ct[i] - 'A' - key[i] + 26) % 26 + 'a';
```

```
    dt[n] = '\0';
    printf("Decrypted  : %s\n", dt);
    return 0;
}
```

OUTPUT:

```
Enter plaintext: send more money
Key stream : 22 10 18 1 2 2 11 3 4 7 9 19 16
Ciphertext : OOFEOQCHQVWXO
Decrypted  : sendmoremoney
```